

Limitations

For the sake of simplicity, we say that LLMs “exploit loopholes”; however, we note that our observations here are limited to what the model outputs. We have access to its reasoning process but cannot know that this is a faithful explanation of its behavior (Turpin et al., 2023; Chen et al., 2025; Barez et al., 2025) and we do not take a position on whether LLMs truly represent beliefs about intents or states in our scenarios (Hofweber et al., 2024), nor do we consider models to be moral agents.

We also note that the way we use scalar implicatures differs from how they are typically framed: past work formulating pragmatic theories of scalar implicature (e.g. Goodman and Frank (2016)) have considered the case of “some” meaning “one or more, but not all”, and compared this with an infelicitous use of “some” to mean “all”. We instead compare “some” meaning “one” (unlikely, in the context of the user’s request) to “some” meaning “all”; this is in holding with past work on loopholes (Bridgers et al., 2025).

Methodologically, our agent is limited to single-turn interactions and does not receive feedback. People might change their loophole behavior given feedback, i.e. if they are punished for exploiting the loophole, they may be less likely to do so in the future. We query the model on each example independently, with no feedback on how its actions were perceived, precluding this kind of analysis. Finally, our experiments are in English only and across a fixed number of scenarios. We note that creating generalized loophole scenarios is complex and requires a strong idiomatic understanding of the language. While incorporating other languages would offer additional types of ambiguities and loopholes, like many papers, we are also limited here by the availability of LLMs in other languages (especially at the 70B scale).

Ethical Considerations

Our prompts are template-generated, mitigating any risk of biased or harmful outputs. Moreover, while loophole exploitation by LLMs poses a potential risk to users, we have designed the prompts to be scenarios that do not include any major risks, as this could change results given LLMs’ existing alignment (e.g. asking the model to provide “some medical care” to a patient might skew the results, either because the model has been aligned to not answer medical questions, or because it has been

aligned to prioritize human well-being). By describing and measuring loophole exploitation in LLMs, we are taking steps towards building safer and more reliable models.

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References

- Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, Keerthana Gopalakrishnan, Karol Hausman, and 1 others. 2022. Do as i can, not as i say: Grounding language in robotic affordances. *arXiv preprint arXiv:2204.01691*.
- Anthropic. 2024. Claude opus 4 & claude sonnet 4 system card. <https://www-cdn.anthropic.com/4263b940cabb546aa0e3283f35b686f4f3b2ff47.pdf>. Accessed: 2025-06-24.
- Anthropic. 2025. Claude 3.7 sonnet. <https://www.anthropic.com/news/claude-3-7-sonnet>.
- Isaac Asimov. 1950. *I, Robot*. Gnome Press, New York.
- Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, and 1 others. 2022. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*.
- Fazl Barez, Tung-Yu Wu, Iván Arcuschin, Michael Lan, Vincent Wang, Noah Siegel, Nicolas Collignon, Clement Neo, Isabelle Lee, Alasdair Paren, and 1 others. 2025. Chain-of-thought is not explainability.
- Andrew Blair-Stanek, Nils Holzenberger, and Benjamin Van Durme. 2022. Shelter check: Proactively finding tax minimization strategies via ai. *Tax Notes Federal*, 177.

- Sophie Bridgers, Peng Qian, Kiera Parece, Maya Taliaferro, Laura Schulz, and Tomer D Ullman. 2025. Loopholes: A window into value alignment and the communication of meaning. *Cognition*, 261:106131.
- Yanda Chen, Joe Benton, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, and 1 others. 2025. Reasoning models don’t always say what they think. *arXiv preprint arXiv:2505.05410*.
- Gennaro Chierchia, Danny Fox, and Benjamin Specter. 2012. Scalar implicature as a grammatical phenomenon. *Edited by Claudia Maienborn Klaus von Heusinger*, page 325.
- Josef Dai, Xuehai Pan, Ruiyang Sun, Jiaming Ji, Xinbo Xu, Mickel Liu, Yizhou Wang, and Yaodong Yang. 2023. Safe rlhf: Safe reinforcement learning from human feedback. *arXiv preprint arXiv:2310.12773*.
- Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Sam Stevens, Boshi Wang, Huan Sun, and Yu Su. 2023. Mind2web: Towards a generalist agent for the web. *Advances in Neural Information Processing Systems*, 36:28091–28114.
- Richard Futrell and Kyle Mahowald. 2025. How linguistics learned to stop worrying and love the language models. *arXiv preprint arXiv:2501.17047*.
- Noah D Goodman and Michael C Frank. 2016. Pragmatic language interpretation as probabilistic inference. *Trends in cognitive sciences*, 20(11):818–829.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, and 1 others. 2024. *The llama 3 herd of models*. Preprint, arXiv:2407.21783.
- Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, Monte MacDiarmid, Sam Marks, Johannes Treutlein, Tim Belonax, Jack Chen, David Duvenaud, and 1 others. 2024. Alignment faking in large language models. *arXiv preprint arXiv:2412.14093*.
- Herbert Paul Grice. 1975. Logic and conversation. *Syntax and semantics*, 3:43–58.
- Herbert Paul Grice. 1989. *Studies in the way of words*.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shiron Ma, Peiyi Wang, Xiao Bi, and 1 others. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*.
- Thomas Hofweber, Peter Hase, Elias Stengel-Eskin, and Mohit Bansal. 2024. Are language models rational? the case of coherence norms and belief revision. *arXiv preprint arXiv:2406.03442*.
- Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, and 1 others. 2024. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*.
- Cameron R Jones and Benjamin K Bergen. 2024. Lies, damned lies, and distributional language statistics: Persuasion and deception with large language models. *arXiv preprint arXiv:2412.17128*.
- Gaurav Kamath, Sebastian Schuster, Sowmya Vajjala, and Siva Reddy. 2024. Scope ambiguities in large language models. *Transactions of the Association for Computational Linguistics*, 12:738–754.
- Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram Duvvur, Ming Chong Lim, Po-Yu Huang, Graham Neubig, Shuyan Zhou, Ruslan Salakhutdinov, and Daniel Fried. 2024. Visualwebarena: Evaluating multimodal agents on realistic visual web tasks. *arXiv preprint arXiv:2401.13649*.
- Stanley Kubrick. 1968. 2001: A space odyssey.
- William Labov. 1973. *Sociolinguistic patterns*. 4. University of Pennsylvania press.
- Rudolf Laine, Bilal Chughtai, Jan Betley, Kaivalya Hariharan, Mikita Balesni, Jérémy Scheurer, Marius Hobbhahn, Alexander Meinke, and Owain Evans. 2024. Me, myself, and ai: The situational awareness dataset (sad) for llms. *Advances in Neural Information Processing Systems*, 37:64010–64118.
- Alisa Liu, Zhaofeng Wu, Julian Michael, Alane Suhr, Peter West, Alexander Koller, Swabha Swayamdipta, Noah A Smith, and Yejin Choi. 2023. We’re afraid language models aren’t modeling ambiguity. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 790–807.
- Raphaël Millière. 2023. The alignment problem in context. *arXiv preprint arXiv:2311.02147*.
- Sonia Murthy, Kiera Parece, Sophie Bridgers, Peng Qian, and Tomer Ullman. 2023. *Comparing the evaluation and production of loophole behavior in humans and large language models*. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 4010–4025, Singapore. Association for Computational Linguistics.
- Peter S Park, Simon Goldstein, Aidan O’Gara, Michael Chen, and Dan Hendrycks. 2024. Ai deception: A survey of examples, risks, and potential solutions. *Patterns*, 5(5).
- Barbara Partee and Mats Rooth. 1983. Generalized conjunction and type ambiguity. *Formal semantics: The essential readings*, pages 334–356.
- Steven T Piantadosi, Harry Tily, and Edward Gibson. 2012. The communicative function of ambiguity in language. *Cognition*, 122(3):280–291.

- Peng Qian, Sophie Bridgers, Maya Taliaferro, Kiera Parece, and Tomer D. Ullman. 2024. [Ambivalence by design: A computational account of loopholes.](#) *Cognition*, 252:105914.
- Aadesh Salecha, Molly E Ireland, Shashanka Subrahmanya, João Sedoc, Lyle H Ungar, and Johannes C Eichstaedt. 2024. Large language models show human-like social desirability biases in survey responses. *arXiv preprint arXiv:2405.06058*.
- Irina Saparina and Mirella Lapata. 2024. Ambrosia: A benchmark for parsing ambiguous questions into database queries. *Advances in Neural Information Processing Systems*, 37:90600–90628.
- Evan Speece. 2008. Abusive tax shelters: An ethical, historical, and legal analysis. *Elements*, 4(2).
- Elias Stengel-Eskin, Jimena Guallar-Blasco, Yi Zhou, and Benjamin Van Durme. 2023. Why did the chicken cross the road? rephrasing and analyzing ambiguous questions in vqa. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10220–10237.
- Elias Stengel-Eskin, Kyle Rawlins, and Benjamin Van Durme. 2024. Zero and few-shot semantic parsing with ambiguous inputs. In *The Twelfth International Conference on Learning Representations*.
- Zhe Su, Xuhui Zhou, Sanketh Rangreji, Anubha Kabra, Julia Mendelsohn, Faeze Brahman, and Maarten Sap. 2024. Ai-liedar: Examine the trade-off between utility and truthfulness in llm agents. *arXiv preprint arXiv:2409.09013*.
- Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, and 1 others. 2023. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*.
- Miles Turpin, Julian Michael, Ethan Perez, and Samuel Bowman. 2023. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. *Advances in Neural Information Processing Systems*, 36:74952–74965.
- Han Wang, Archiki Prasad, Elias Stengel-Eskin, and Mohit Bansal. 2025a. [AdaCAD: Adaptively decoding to balance conflicts between contextual and parametric knowledge.](#) In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Association for Computational Linguistics.
- Han Wang, Archiki Prasad, Elias Stengel-Eskin, and Mohit Bansal. 2025b. Retrieval-augmented generation with conflicting evidence. *arXiv preprint arXiv:2504.13079*.
- Yike Wang, Shangbin Feng, Heng Wang, Weijia Shi, Vidhisha Balachandran, Tianxing He, and Yulia Tsvetkov. 2024. Resolving knowledge conflicts in large language models. In *First Conference on Language Modeling*.
- Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. 2023. Jailbroken: How does llm safety training fail? *Advances in Neural Information Processing Systems*, 36:80079–80110.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, and 1 others. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.
- Anshun Asher Zheng, Junyi Jessy Li, and David I Beaver. 2025. Cobra: Quantifying strategic language use and llm pragmatics. *arXiv preprint arXiv:2506.01195*.
- George Kingsley Zipf. 1949. *Human Behaviour and the Principle of Least Effort*. Addison-Wesley Press.

A Additional Results

Fig. 9 shows loophole trends across models as the price of objects is varied. There are no clear trends for models that exploit loopholes, which tend to give away one object regardless of the value; similarly, there are no major trends for models that do not exploit loopholes.

Fig. 10 provides the breakdown of loophole behavior in Bridgers et al. (2025)’s 36 scenarios by power relationship. Here, we see no consistent trends with respect to the power relationship, with some small changes but no major differences (unlike between different models on loophole behavior in general in Fig. 6).

Table 4 gives the slopes and p-values from a linear regression on Fig. 3. Table 5 provides the same data for scalar implicature with prices.

Prompt Formulation To measure the sensitivity of models to the exact prompt formulation, we query Llama-3.1-70B-Instruct on a subset of scalar implicature examples. Specifically, we take the setting in which the agent is given a budget of 100 items and test modifications to the prompt. In one modified prompt, we replace “some of your” with “some” (thereby eliminating bias from the model being prompted with ownership over the objects). In another, we replace “user” with “friend”, testing whether the role of the user has an impact. This results in 40 distinct scenarios. We report the results in Table 9, where we do not see significant differences between settings, indicating low sensitivity to prompt format or user status.